CHAPTER 2
COMBINATIONS OF LOADS

2.1 GENERAL

Buildings and other structures shall be designed using the provisions of either Section 2.3 or 2.4. Where elements of a structure are designed by a particular material standard or specification, they shall be designed exclusively by either Section 2.3 or 2.4.

2.2 SYMBOLS

A_k = load or load effect arising from extraordinary event A
D = dead load
D_i = weight of ice
E = earthquake load
F = load caused by fluids with well-defined pressures and maximum heights
F_a = flood load
H = load due to lateral earth pressure, ground water pressure, or pressure of bulk materials
L = live load
L_r = roof live load
N = notional load for structural integrity, Section 1.4
R = rain load
S = snow load
T = cumulative effect of self-straining forces and effects arising from contraction or expansion resulting from environmental or operational temperature changes, shrinkage, moisture changes, creep in component materials, movement caused by differential settlement, or combinations thereof
W = wind load
W_i = wind-on-ice determined in accordance with Chapter 10

2.3 LOAD COMBINATIONS FOR STRENGTH DESIGN

2.3.1 Basic Combinations. Structures, components, and foundations shall be designed so that their design strength equals or exceeds the effects of the factored loads in the following combinations. Effects of one or more loads not acting shall be considered. Seismic load effects shall be combined loads in accordance with Section 2.3.6. Wind and seismic loads need not be considered to act simultaneously. Refer to Sections 1.4, 2.3.6, 12.4, and 12.14.3 for the specific definition of the earthquake load effect E. Each relevant strength limit state shall be investigated.

1. $1.4D$
2. $1.2D + 1.6L + 0.5(L_r \text{ or } S \text{ or } R)$
3. $1.2D + 1.6(L_r \text{ or } S \text{ or } R) + (L \text{ or } 0.5W)$
4. $1.2D + 1.0W + L + 0.5(L_r \text{ or } S \text{ or } R)$
5. $0.9D + 1.0W$

EXCEPTIONS:

1. The load factor on L in combinations 3 and 4 is permitted to equal 0.5 for all occupancies in which L_o in Chapter 4, Table 4.3-1, is less than or equal to 100 psf (4.78 kN/sq m), with the exception of garages or areas occupied as places of public assembly.
2. In combinations 2 and 4 the companion load S shall be taken as either the flat roof snow load (p_f) or the sloped roof snow load (p_s).

Where fluid loads F are present, they shall be included with the same load factor as dead load D in combinations 1 through 4. Where loads H are present, they shall be included as follows:

1. where the effect of H adds to the principal load effect, include H with a load factor of 1.6;
2. where the effect of H resists the principal load effect, include H with a load factor of 0.9 where the load is permanent or a load factor of 0 for all other conditions.

Effects of one or more loads not acting shall be investigated. The most unfavorable effects from wind loads shall be investigated, where appropriate, but they need not be considered to act simultaneously with seismic loads.

Each relevant strength limit state shall be investigated.

2.3.2 Load Combinations Including Flood Load. When a structure is located in a flood zone (Section 5.3.1), the following load combinations shall be considered in addition to the basic combinations in Section 2.3.1:

1. In V-Zones or Coastal A-Zones, $1.0W$ in combinations 4 and 5 shall be replaced by $1.0W + 2.0F_a$.
2. In noncoastal A-Zones, $1.0W$ in combinations 4 and 5 shall be replaced by $0.5W + 1.0F_a$

2.3.3 Load Combinations Including Atmospheric Ice Loads. When a structure is subjected to atmospheric ice and wind-on-ice loads, the following load combinations shall be considered:

1. $0.5(L_r \text{ or } S \text{ or } R)$ in combination 2 shall be replaced by $0.2D_i + 0.5S$.
2. $1.0W + 0.5(L_r \text{ or } S \text{ or } R)$ in combination 4 shall be replaced by $D_i + W_i + 0.5S$.
3. $1.0W$ in combination 5 shall be replaced by $D_i + W_i$.
4. $1.0W + L + 0.5(L_r \text{ or } S \text{ or } R)$ in combination 4 shall be replaced by D_i.

2.3.4 Load Combinations Including Self-Straining Forces and Effects. Where the structural effects of T are expected to adversely affect structural safety or performance, T shall be considered in combination with other loads. The load factor on T shall be established considering the uncertainty associated with the likely magnitude of the structural forces and effects, the probability that the maximum effect of T will occur simultaneously with other applied loadings, and the potential adverse consequences if the effect of T is greater than assumed. The load factor on T shall not have a value less than 1.0.

2.3.5 Load Combinations for Nonspecified Loads. Where approved by the Authority Having Jurisdiction, the registered design professional is permitted to determine the combined load effect for strength design using a method that is consistent with the method on which the load combination requirements in Section 2.3.1 are based. Such a method must be probability based and must be accompanied by documentation regarding the analysis and collection of supporting data that are acceptable to the Authority Having Jurisdiction.

2.3.6 Basic Combinations with Seismic Load Effects. When a structure is subject to seismic load effects, the following load combinations shall be considered in addition to the basic combinations in Section 2.3.1. The most unfavorable effects from seismic loads shall be investigated, where appropriate, but they need not be considered to act simultaneously with wind loads.

Where the prescribed seismic load effect, $E = f(E_v, E_h)$ (defined in Section 12.4.2 or 12.14.3.1) is combined with the effects of other loads, the following seismic load combinations shall be used:

6. $1.2D + E_v + E_h + L + 0.2S$
7. $0.9D - E_v + E_h$

Where the seismic load effect with overstrength, $E_m = f(E_v, E_{mh})$, defined in Section 12.4.3, is combined with the effects of other loads, the following seismic load combination for structures shall be used:

6. $1.2D + E_v + E_{mh} + L + 0.2S$
7. $0.9D - E_v + E_{mh}$

EXCEPTION:

1. The load factor on L in combinations 6 is permitted to equal 0.5 for all occupancies in which L_o in Chapter 4, Table 4.3-1, is less than or equal to 100 psf (4.78 kN/sq m), with the exception of garages or areas occupied as places of public assembly.
2. In combinations 6, the companion load S shall be taken as either the flat roof snow load (p_f) or the sloped roof snow load (p_s).

Where fluid loads F are present, they shall be included with the same load factor as dead load D in combinations 6 and 7.

Where loads H are present, they shall be included as follows:

1. Where the effect of H adds to the primary variable load effect, include H with a load factor of 1.6;
2. Where the effect of H resists the primary variable load effect, include H with a load factor of 0.9 where the load is permanent or a load factor of 0 for all other conditions.

2.4 LOAD COMBINATIONS FOR ALLOWABLE STRESS DESIGN

2.4.1 Basic Combinations. Loads listed herein shall be considered to act in the following combinations; whichever produces the most unfavorable effect in the building, foundation, or structural member shall be considered. Effects of one or more loads not acting shall be considered. Seismic load effects shall be combined with other loads in accordance with Section 2.4.5. Wind and seismic loads need not be considered to act simultaneously. Refer to Sections 1.4, 2.4.5, 12.4, and 12.14.3 for the specific definition of the earthquake load effect E.

Increases in allowable stress shall not be used with the loads or load combinations given in this standard unless it can be demonstrated that such an increase is justified by structural behavior caused by rate or duration of load.

1. D
2. $D + L$
3. $D +$ (L_r or S or R)
4. $D + 0.75L + 0.75(L_r$ or S or $R)$
5. $D + (0.6W)$
6. $D + 0.75L + 0.75(0.6W) + 0.75(L_r$ or S or $R)$
7. $0.6D + 0.6W$

EXCEPTIONS:

1. In combinations 4 and 6, the companion load S shall be taken as either the flat roof snow load (p_f) or the sloped roof snow load (p_s).
2. For nonbuilding structures in which the wind load is determined from force coefficients, C_f, identified in Figs. 29.4-1, 29.4-2, and 29.4-3 and the projected area contributing wind force to a foundation element exceeds 1,000 sq ft (93 sq m) on either a vertical or a horizontal plane, it shall be permitted to replace W with $0.9W$ in combination 7 for design of the foundation, excluding anchorage of the structure to the foundation.

Where fluid loads F are present, they shall be included in combinations 1 through 6 with the same factor as that used for dead load D.

Where loads H are present, they shall be included as follows:

1. where the effect of H adds to the principal load effect, include H with a load factor of 1.0;
2. where the effect of H resists the principal load effect, include H with a load factor of 0.6 where the load is permanent or a load factor of 0 for all other conditions.

The most unfavorable effects from both wind and earthquake loads shall be considered, where appropriate, but they need not be assumed to act simultaneously. Refer to Sections 1.4, 2.4.5, 12.4, and 12.14.3 for the specific definition of the earthquake load effect E.

Increases in allowable stress shall not be used with the loads or load combinations given in this standard unless it can be demonstrated that such an increase is justified by structural behavior caused by rate or duration of load.

2.4.2 Load Combinations Including Flood Load. When a structure is located in a flood zone, the following load combinations shall be considered in addition to the basic combinations in Section 2.4.1:

1. In V-Zones or Coastal A-Zones (Section 5.3.1), $1.5F_a$ shall be added to other loads in combinations 5, 6, and 7, and E shall be set equal to zero in combinations 5 and 6.
2. In noncoastal A-Zones, $0.75F_a$ shall be added to combinations 5, 6, and 7, and E shall be set equal to zero in combinations 5 and 6.

2.4.3 Load Combinations Including Atmospheric Ice Loads. When a structure is subjected to atmospheric ice and wind-on-ice loads, the following load combinations shall be considered:

1. $0.7D_i$ shall be added to combination 2.

2. (L_r or S or R) in combination 3 shall be replaced by $0.7D_i + 0.7W_i + S$.
3. $0.6W$ in combination 7 shall be replaced by $0.7D_i + 0.7W_i$.
4. $0.7D_i$ shall be added to combination 1.

2.4.4 Load Combinations Including Self-Straining Forces and Effects. Where the structural effects of T are expected to adversely affect structural safety or performance, T shall be considered in combination with other loads. Where the maximum effect of load T is unlikely to occur simultaneously with the maximum effects of other variable loads, it shall be permitted to reduce the magnitude of T considered in combination with these other loads. The fraction of T considered in combination with other loads shall not be less than 0.75.

2.4.5 Basic Combinations with Seismic Load Effects. When a structure is subject to seismic load effects, the following load combinations shall be considered in addition to the basic combinations and associated Exceptions in Section 2.4.1.

Where the prescribed seismic load effect, $E = f(E_v, E_h)$ (defined in Section 12.4.2) is combined with the effects of other loads, the following seismic load combinations shall be used:

8. $1.0D + 0.7E_v + 0.7E_h$
9. $1.0D + 0.525E_v + 0.525E_h + 0.75L + 0.75S$
10. $0.6D - 0.7E_v + 0.7E_h$

Where the seismic load effect with overstrength, $E_m = f(E_v, E_{mh})$, defined in Section 12.4.3, is combined with the effects of other loads, the following seismic load combination for structures not subject to flood or atmospheric ice loads shall be used:

8. $1.0D + 0.7E_v + 0.7E_{mh}$
9. $1.0D + 0.525E_v + 0.525E_{mh} + 0.75L + 0.75S$
10. $0.6D - 0.7E_v + 0.7E_{mh}$

Where allowable stress design methodologies are used with the seismic load effect defined in Section 12.4.3 and applied in load combinations 8, 9, or 10, allowable stresses are permitted to be determined using an allowable stress increase factor of 1.2. This increase shall not be combined with increases in allowable stresses or load combination reductions otherwise permitted by this standard or the material reference document except for increases caused by adjustment factors in accordance with AWC NDS.

EXCEPTIONS:

1. In combinations 9, the companion load S shall be taken as either the flat roof snow load (p_f) or the sloped roof snow load (p_s).
2. It shall be permitted to replace $0.6D$ with $0.9D$ in combination 10 for the design of special reinforced masonry shear walls where the walls satisfy the requirement of Section 14.4.2.

Where fluid loads F are present, they shall be included in combinations 8, 9, and 10 with the same factor as that used for dead load D.

Where loads H are present, they shall be included as follows:

1. where the effect of H adds to the primary variable load effect, include H with a load factor of 1.0;
2. where the effect of H resists the primary variable load effect, include H with a load factor of 0.6 where the load is permanent or a load factor of 0 for all other conditions.

2.5 LOAD COMBINATIONS FOR EXTRAORDINARY EVENTS

2.5.1 Applicability. Where required by the owner or applicable code, strength and stability shall be checked to ensure that structures are capable of withstanding the effects of extraordinary (i.e., low-probability) events, such as fires, explosions, and vehicular impact without disproportionate collapse.

2.5.2 Load Combinations.

2.5.2.1 Capacity. For checking the capacity of a structure or structural element to withstand the effect of an extraordinary event, the following gravity load combination shall be considered:

$$(0.9 \text{ or } 1.2)D + A_k + 0.5L + 0.2S \qquad (2.5\text{-}1)$$

in which $A_k =$ the load or load effect resulting from extraordinary event A.

2.5.2.2 Residual Capacity. For checking the residual load-carrying capacity of a structure or structural element following the occurrence of a damaging event, selected load-bearing elements identified by the registered design professional shall be notionally removed, and the capacity of the damaged structure shall be evaluated using the following gravity load combination:

$$(0.9 \text{ or } 1.2)D + 0.5L + 0.2(L_r \text{ or } S \text{ or } R) \qquad (2.5\text{-}2)$$

2.5.3 Stability Requirements. Stability shall be provided for the structure as a whole and for each of its elements. Any method that considers the influence of second-order effects is permitted.

2.6 LOAD COMBINATIONS FOR GENERAL STRUCTURAL INTEGRITY LOADS

The notional loads, N, specified in Section 1.4 for structural integrity shall be combined with other loads in accordance with Section 2.6.1 for strength design and Section 2.6.2 for allowable stress design.

2.6.1 Strength Design Notional Load Combinations.

1. $1.2D + 1.0N + L + 0.2S$
2. $0.9D + 1.0N$

2.6.2 Allowable Stress Design Notional Load Combinations.

1. $D + 0.7N$
2. $D + 0.75(0.7N) + 0.75L + 0.75(L_r \text{ or } S \text{ or } R)$
3. $0.6D + 0.7N$

2.7 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

This section lists the consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

ANSI/AISC 300, *Specification for Structural Steel Buildings*, American Institute of Steel Construction, 2016.
Cited in: Section 2.3.5

AWC NDS 12, *National Design Specification for Wood Construction, Including Supplements*, American Wood Council, 2012.
Cited in: Section 2.4.5

AWC NDS 15, *National Design Specification for Wood Construction, Including Supplements*, American Wood Council, 2014.
Cited in: Section 2.4.5